MPU-6886 Gyroscope Bias Drift Analysis

Addendum to MPU6886_ThermalDrift_Report.pdf

Internal R&D Report

Test ID	tel_94
Firmware	v1.2.2-diag-kgs
Data frame	chip-frame (K_gs off, rotate_3d off for gyro)
Duration (lap=1)	2100s (35.0 min)
Total samples	104,984
Sample rate	50 Hz
Temperature range	37.6 – 39.6°C
Date generated	2026-04-11

Key finding:

The MPU-6886 exhibits significant non-thermal bias drift on gx and gy

(gx: -0.0762 °/s/min, gy: -0.0301 °/s/min)

at near-constant temperature (37.6 – 39.6°C)

6-Face Static Test — Full Timeline (lap=1, chip-frame)
Faces
Z_up
Z_down
X_up
X_down
Y_up
Y_down
Accelerometer norm — transitions visible as spikes
|a| (g)
Gyroscope chip-frame — monotonic drift visible on gx (red)
Gyro (°/s)
gx
gy
gz
Accelerometer chip-frame — dominant axis confirms face orientation
Accel (g)
ax
ay
az
IMU temperature — nearly flat throughout the test
Temp (°C)
Time (s)

Face 1: Z_up — Y-chip up (+0.997g)

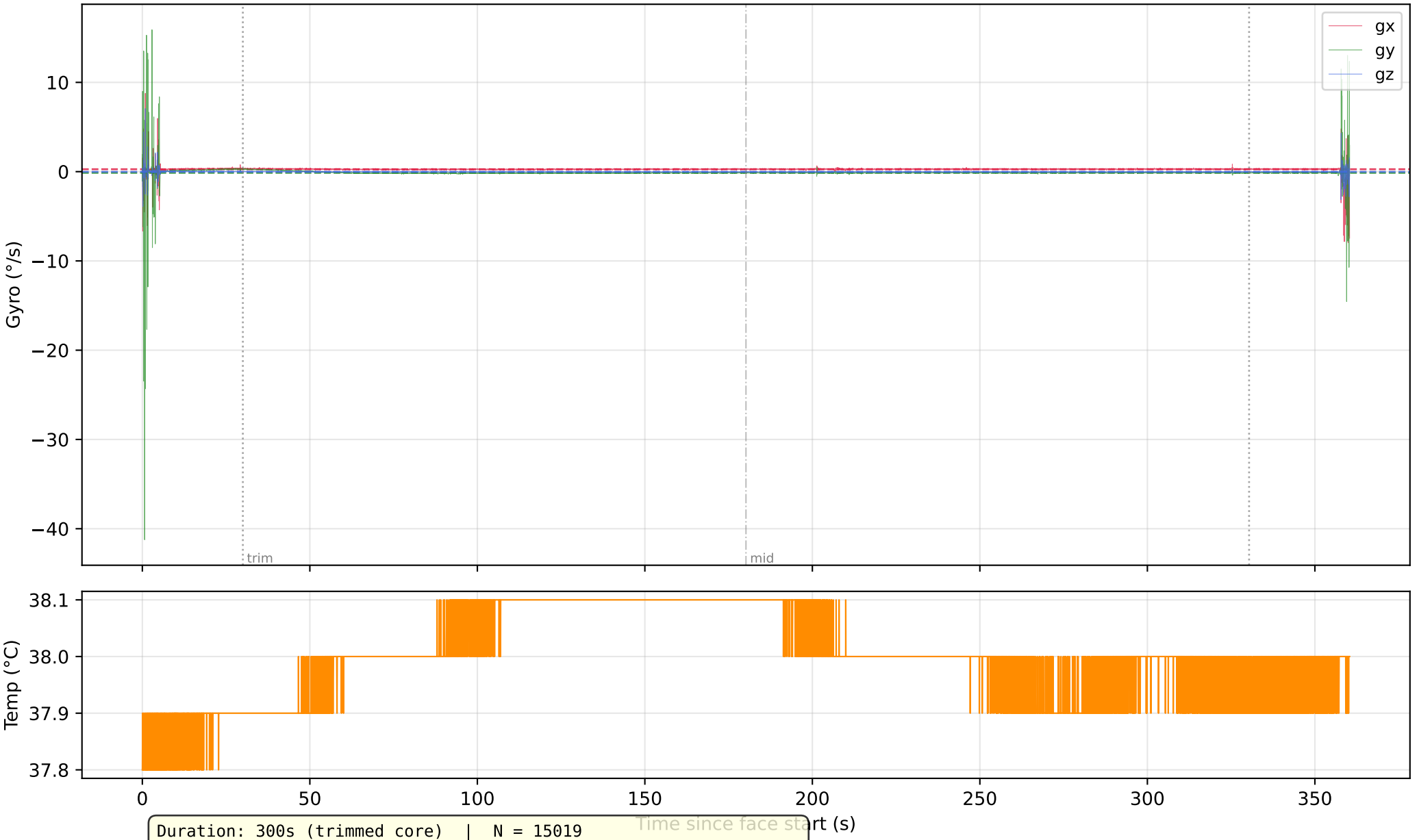
```
Duration: 275s (trimmed core)  |  N = 13750

Trimmed means:   gx = +0.4253    gy = -0.0619    gz = +0.0152 °/s
Std:             gx = 0.0483     gy = 0.0392     gz = 0.0327 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = +0.0269 °/s   Δgy = -0.0263 °/s   Δgz = -0.0081 °/s

Temperature: 37.9 → 37.9°C  (ΔT = +0.00°C)
```

Face 2: Z_down — Y-chip down (-1.005g)



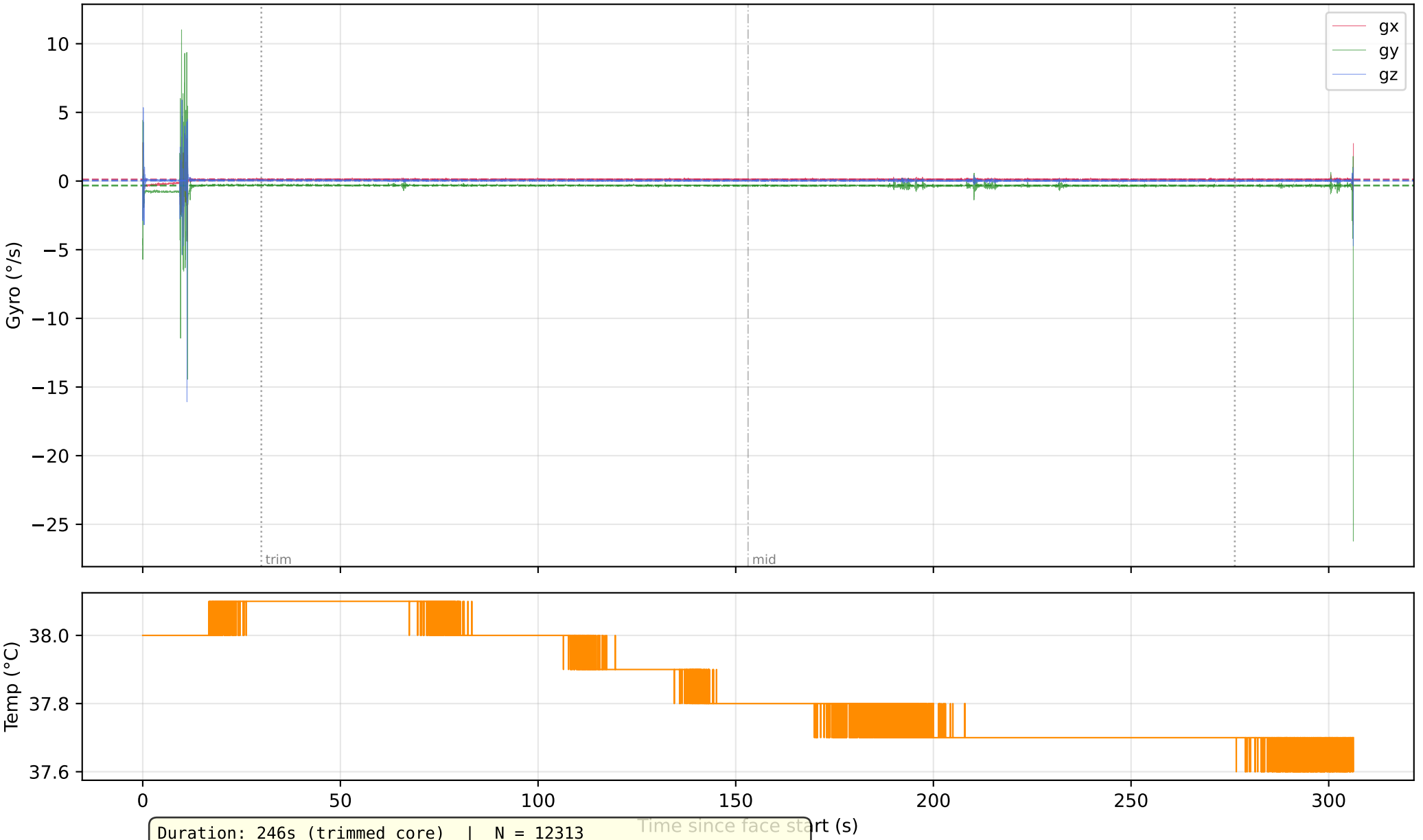

```
Duration: 300s (trimmed core)  |  N = 15019

Trimmed means:   gx = +0.2732    gy = -0.1353    gz = -0.0187 °/s
Std:             gx = 0.0417     gy = 0.0912     gz = 0.0262 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = +0.0133 °/s   Δgy = -0.0008 °/s   Δgz = -0.0064 °/s

Temperature: 37.9 → 37.9°C  (ΔT = +0.00°C)
```

Face 3: X_up — Z-chip up (+1.015g)



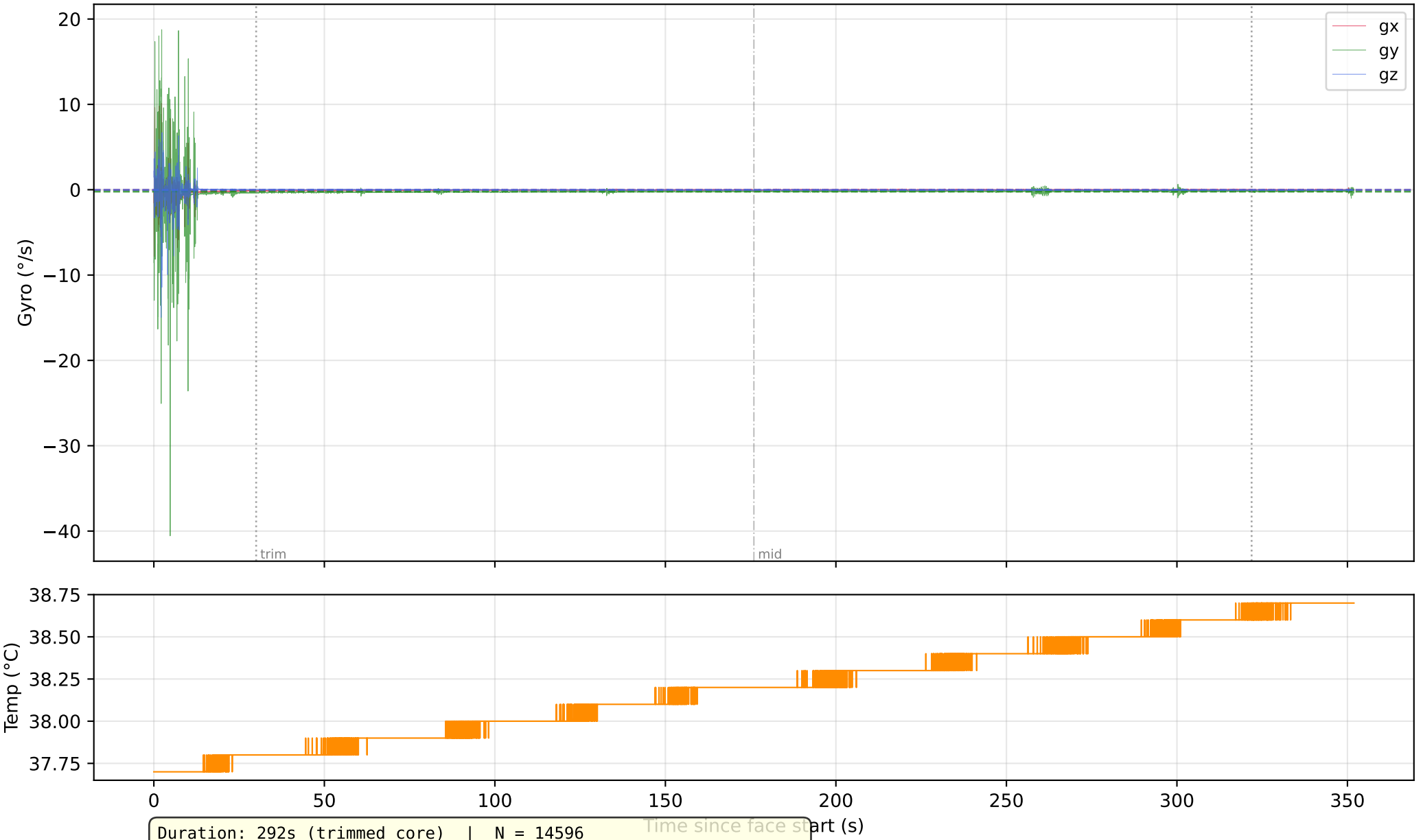

```
Duration: 246s (trimmed core)  |  N = 12313

Trimmed means:   gx = +0.1090    gy = -0.3259    gz = +0.0336 °/s
Std:             gx = 0.0370     gy = 0.0555     gz = 0.0327 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = -0.0062 °/s   Δgy = -0.0257 °/s   Δgz = -0.0144 °/s

Temperature: 38.1 → 37.7°C  (ΔT = -0.40°C)
```

Face 4: X_down — Z-chip down (-0.987g)




```
Duration: 292s (trimmed core)  |  N = 14596

Trimmed means:   gx = -0.0353    gy = -0.2218    gz = -0.0197 °/s
Std:             gx = 0.0429     gy = 0.0700     gz = 0.0275 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = +0.0307 °/s   Δgy = +0.0650 °/s   Δgz = -0.0136 °/s

Temperature: 37.8 → 38.6°C  (ΔT = +0.80°C)
```

Face 5: Y_up — X-chip down (-0.992g)

```
Duration: 275s (trimmed core)  |  N = 13769

Trimmed means:   gx = -0.8220    gy = -0.3615    gz = +0.0471 °/s
Std:             gx = 0.0785     gy = 0.1149     gz = 0.0254 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = +0.1023 °/s   Δgy = +0.1668 °/s   Δgz = -0.0021 °/s

Temperature: 39.2 → 39.4°C  (ΔT = +0.20°C)
```

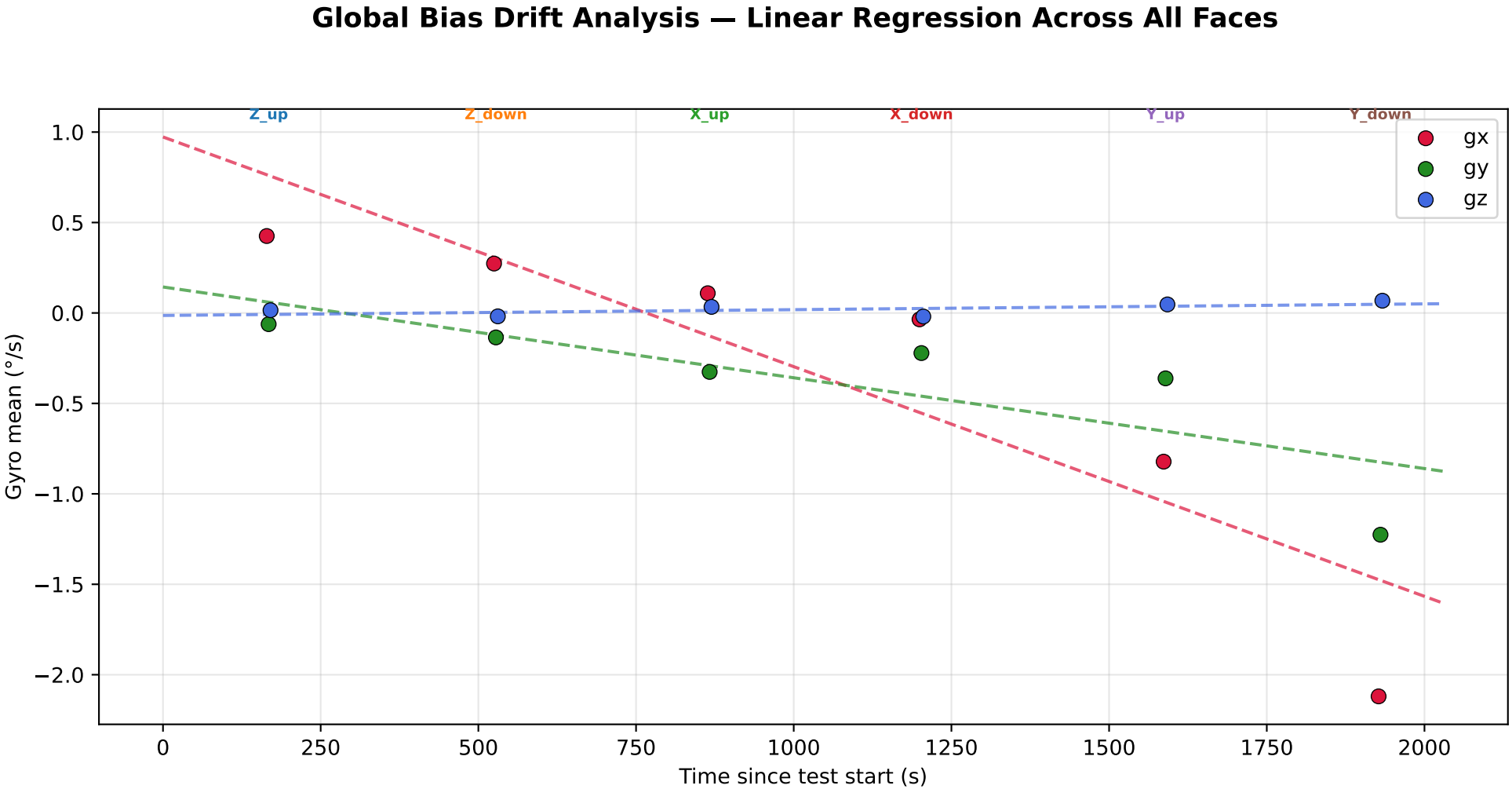
Face 6: Y_down — X-chip up (+1.002g)

```
Duration: 261s (trimmed core)  |  N = 13049

Trimmed means:   gx = -2.1196    gy = -1.2260    gz = +0.0678 °/s
Std:             gx = 0.1293     gy = 0.0899     gz = 0.0306 °/s

Intra-face drift (2nd half - 1st half):
  Δgx = +0.2002 °/s   Δgy = +0.1326 °/s   Δgz = -0.0028 °/s

Temperature: 39.4 → 39.0°C  (ΔT = -0.40°C)
```

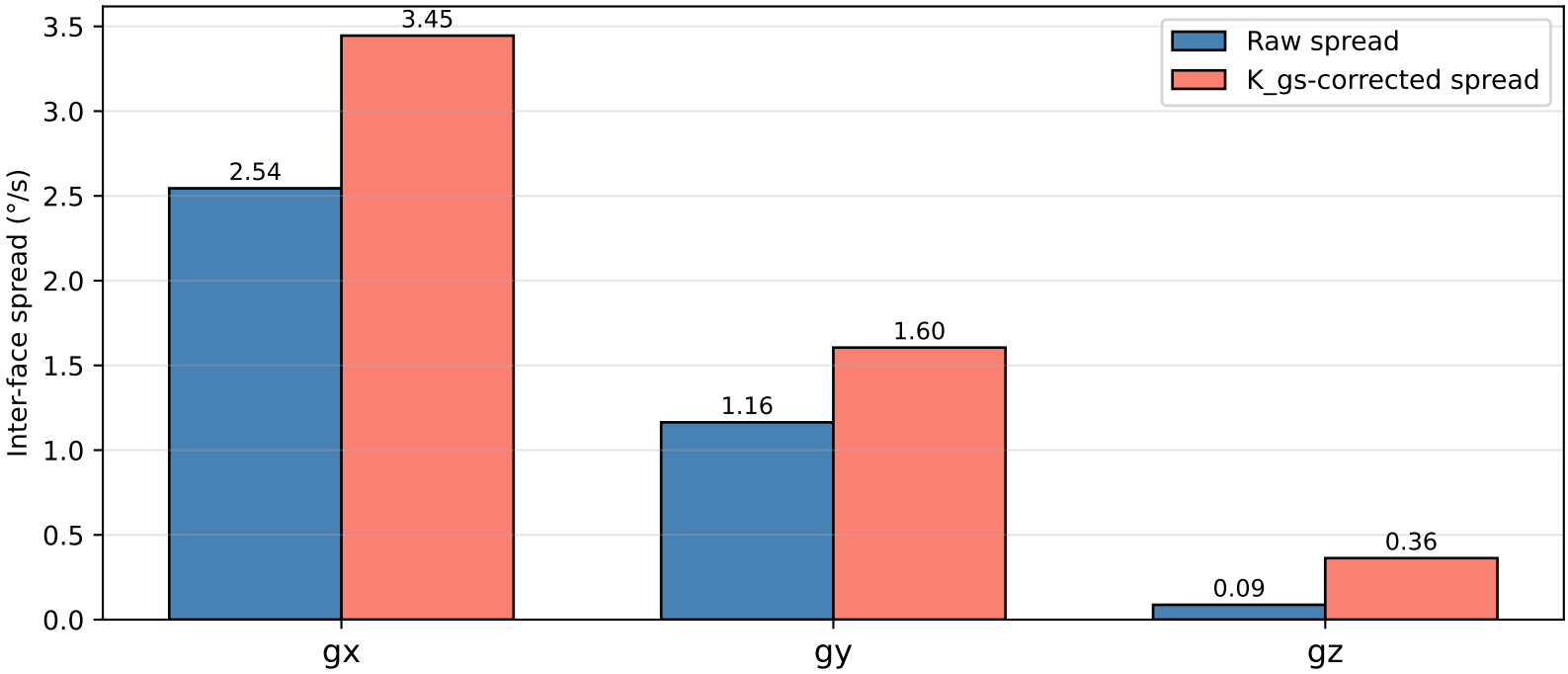


Global Drift Regression (all static samples concatenated)

Axis	Slope	R^2	Total drift	Duration
gx	-0.0762 °/s/min	0.7750	-2.5787 °/s	2031 s
gy	-0.0301 °/s/min	0.5954	-1.0193 °/s	2031 s
gz	+0.0019 °/s/min	0.1943	+0.0648 °/s	2031 s

gx and gy show strong monotonic drift independent of temperature.
gz is stable — critical for ESKF heading estimation.

K_gs Validation — Offline Application on Independent Data



Spread Comparison: Raw (no K_gs) vs K_gs-corrected

Axis	Raw spread	Corrected	Change	Verdict
gx	2.5449 °/s	3.4455 °/s	+35.4%	WORSE
gy	1.1641 °/s	1.6050 °/s	+37.9%	WORSE
gz	0.0875 °/s	0.3629 °/s	+314.8%	WORSE

K_gs correction INCREASES spread on all axes.
Root cause: the inter-face variation is dominated by temporal bias drift, not G-sensitivity. Faces were tested sequentially, so time and orientation are confounded.

Decision: K_gs removed from firmware in v1.3.0.

Comparison with Thermal Drift Report

Previous Finding (MPU6886_ThermalDrift_Report.pdf)

- Gyroscope bias drift was attributed primarily to thermal sensitivity.
- The thermal sensitivity dG/dT was found to be non-repeatable between sessions on GX and GY (inter-run σ ~ 0.10 °/s/°C).
- LUT and Poly-3 compensation both failed: 44% of samples fell in the "unusable" category on GX even after correction.
- Conclusion: no fixed compensation model (LUT or polynomial) can address the non-deterministic drift shape. ZARU is required.

New Finding (this report, tel_94 test)

- Temperature was nearly flat during the test (37.6 – 39.6°C, ΔT < 2°C).
- Despite flat temperature, significant bias drift was observed:

```
gx: -0.0762 °/s/min  (total -2.58 °/s over 2031s)
gy: -0.0301 °/s/min  (total -1.02 °/s over 2031s)
gz: +0.0019 °/s/min  (stable)
```

- The drift is monotonic over 35 minutes – consistent with electronic bias instability (stress relaxation, charge trapping), not thermal.
- K_gs G-sensitivity correction worsened the inter-face spread, confirming that temporal drift dominates over gravity sensitivity.

Reconciliation

Both effects coexist in the MPU-6886:

1. Thermal sensitivity – real, observed in the heating tests.
   The bias changes when temperature changes.

2. Electronic drift – newly quantified in this test.
   The bias changes over time even at constant temperature.

The thermal drift report could not separate these two effects because temperature and time were monotonically correlated in all 7 test runs (continuous heating). The "inter-run shape variability" attributed to non-repeatable thermal sensitivity was likely contaminated by the electronic drift component.

ZARU addresses both: it periodically resets the bias estimate at stationarity, regardless of whether the drift source is thermal or electronic. No firmware change required.

Conclusions

1. GX has the largest non-thermal drift
   Rate: -0.0762 °/s/min (4.6 °/s/hour).
   Over a 10-minute driving stint without stationarity, this accumulates ~0.76 °/s of uncorrected bias error.

2. GY drift is moderate
   Rate: -0.0301 °/s/min (1.8 °/s/hour).
   Less critical: gy (pitch rate) does not feed the 2D ESKF heading.

3. GZ is stable
   Rate: +0.0019 °/s/min (0.11 °/s/hour).
   Excellent for the ESKF, which uses gz for heading estimation.
   Note: in vehicle frame, gz maps from a mix of chip-frame axes through rotate_3d. Chip-frame gx/gy drift can contaminate vehicle-frame gz depending on mounting angle, but the effect is small for near-level mounting (sin of small angles).

4. K_gs correction is counterproductive
   Spread after K_gs: gx +35%, gy +38%, gz +315% (all worse).
   The inter-face variation is dominated by temporal drift, not gravity sensitivity. K_gs removed from firmware in v1.3.0.

5. ZARU remains the primary defense
   The 3-axis ZARU (v1.2.1) corrects both thermal and electronic drift in real-time by measuring stationarity. The current 50-sample (1s) window is adequate. No firmware change required.

6. Recommendation for future calibration tests
   Randomize face order (not sequential) to decorrelate temporal drift from gravity orientation. This would allow cleaner separation of G-sensitivity and bias drift effects, if K_gs is ever revisited.

Test conditions: tel_94.bin, firmware v1.2.2-diag-kgs
K_gs disabled, rotate_3d bypassed on gyro
raw_ax/ay/az = chip-frame accel (not linear accel)
Temperature: 37.6 – 39.6°C (near-constant)
Duration: 2031s of static data across 6 faces